Givens

$Z_o = 50\Omega, \qquad f = 9.5GHz$

$Copper\ thickness = 1\ oz/ft^2 = 34\ \mu m$

Substrate: Rogers RT/Duroid 5880

$h = 787.41\ \mu m, \qquad \varepsilon_r = 2.20 \pm 0.04, \qquad tan\delta = 0.0009\ @\ 10\ GHz$

npn type BJT BFG410W fabricated by NXP semiconductors

$I_c = 6mA, \qquad V_{ce} = 2V$

S parameters @$9.5GHz$ are Extracted from Spar_BFG410W_commonemitter2V6mA file

$$[S] = \begin{bmatrix} 0.608\angle 34.757° & 0.151\angle -52.987° \\ 1.603\angle -81.484° & 0.238\angle 49.648° \end{bmatrix}$$

Matching Networks Microstrip TL — radial stub and $\lambda/4$ transmission line $\left(\lambda = \lambda_g = \frac{\lambda_o}{\sqrt{\varepsilon_{eff}}}\right)$

Specs

- Stability: stable over the entire useful bandwidth of the device
- $G = G_{T,max}$ at the design frequency
- $S11|_{amp}, S22|_{amp}$: minimize
- Only open-circuited shunt stubs should be used
- BW: maximize, based on a narrowband design (The pair of networks that are to be used in the end should be selected such that the bandwidth of the amplifier is maximized)
- Amplifier must be DC-isolated by blocking capacitors
- Layout has space between the launcher microstrip lines and the surface mount capacitors
- Layout must take into consideration the physical dimensions of the transistor, capacitors, and resistors. The capacitors and resistors are surface-mounted ones. You can use the dimensions of the 0603 package for the capacitors and resistors.

Stability

Rollet's condition and μ-stability test

A device will be unconditionally stable if:

$$\left.\begin{array}{l} \circ\ K = \dfrac{1-|S_{11}|^2-|S_{22}|^2+|\Delta|^2}{2|S_{12}S_{21}|} > 1 \\ \circ\ |\Delta| = |S_{11}S_{22} - S_{12}S_{21}| < 1 \end{array}\right\} \text{Rollet's condition}$$

- $\mu = \dfrac{1-|S_{11}|^2}{|S_{22}-S_{11}^{*}\Delta|+|S_{21}S_{12}|} > 1$

$$\Delta = 0.366\angle 59.891^{\circ}$$

$$|\Delta| = 0.366 < 1$$

$$K = 1.462 > 1$$

The device is unconditionally stable @9.5GHz

Another Solution

$$\mu = 1.849 > 1$$

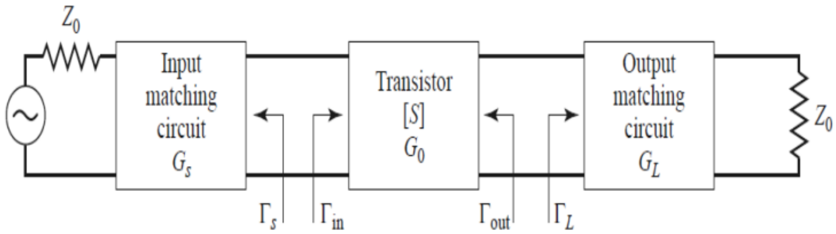
The device is unconditionally stable @9.5GHz

Gain

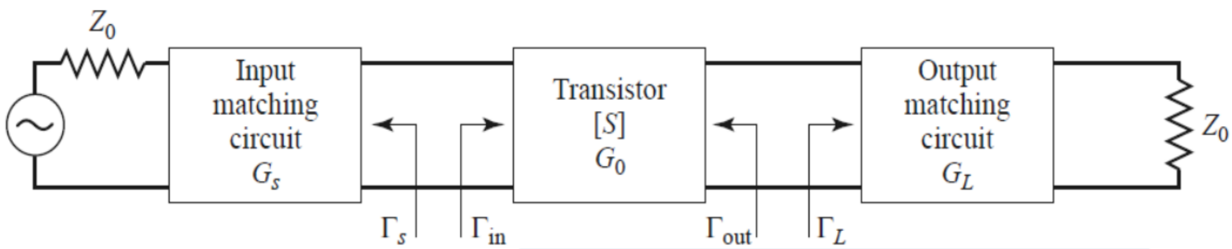
Gain equations

- **The transducer power gain is:**
  - $G_T = \frac{P_L}{P_{AVS}} = \frac{(1-|\Gamma_S|^2)}{|1-\Gamma_{in}\Gamma_S|^2}|S_{21}|^2\frac{(1-|\Gamma_L|^2)}{|1-S_{22}\Gamma_L|^2}$
  - $G_T = \frac{P_L}{P_{AVS}} = \frac{(1-|\Gamma_S|^2)}{|1-S_{11}\Gamma_S|^2}|S_{21}|^2\frac{(1-|\Gamma_L|^2)}{|1-\Gamma_{out}\Gamma_L|^2}$

Maximum gain amplifier design



Maximum gain design (conjugate matching)



- $\Gamma_{in} = \Gamma_S^*$ **and** $\Gamma_{out} = \Gamma_L^*$
- $\Gamma_S^* = S_{11} + \frac{S_{12}S_{21}\Gamma_L}{1-S_{22}\Gamma_L}$
- $\Gamma_L^* = S_{22} + \frac{S_{12}S_{21}\Gamma_S}{1-S_{11}\Gamma_S}$
- $\Gamma_S = \frac{B_1 \pm \sqrt{B_1^2-4|C_1|^2}}{2C_1}$
- $\Gamma_L = \frac{B_2 \pm \sqrt{B_2^2-4|C_2|^2}}{2C_2}$

$$G_{Tmax} = G_{S_{max}} G_0 G_{L_{max}}$$

$$G_{Tmax} = \frac{1}{(1-|\Gamma_S|^2)}|S_{21}|^2\frac{(1-|\Gamma_L|^2)}{|1-S_{22}\Gamma_L|^2}$$

$$G_{T_{max}} = \frac{|S_{21}|}{|S_{12}|}\left(K-\sqrt{K^2-1}\right)$$

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2$$

$$C_1 = S_{11} - \Delta S_{22}^*$$
$$C_2 = S_{22} - \Delta S_{11}^*$$

Law4

$$G_{T,max} = 4.199 = 6.231\ dB$$

Another Solution

$$B_1 = 1.179$$

$$B_2 = 0.553$$

$$C_1 = 0.53\angle 38.669°$$

$$C_2 = 0.099\angle 118.652°$$

$$\Gamma_S = \cancel{1.599\angle -38.669°\ (rejected)}, \qquad 0.625\angle -38.669°$$

$$\Gamma_L = \cancel{5.404\angle -118.652°\ (rejected)}, \qquad 0.185\angle -118.652°$$

$$\Gamma_{in} = 0.625\angle 38.669°$$

$$\Gamma_{out} = 0.185\angle 118.652°$$

Law1

$$G_T = 4.199 = 6.231dB$$

Law2

$$G_T = 4.199 = 6.231dB$$

Law3

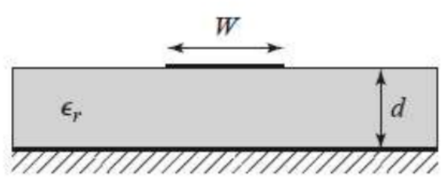
$$G_T = 4.199 = 6.231dB$$

Microstrip TL Design

Microstrip transmission lines: $\frac{W}{d}$

For a given characteristic impedance Z_o and dielectric constant ϵ_r, the *W/d* ratio can be found as:

- $$\frac{W}{d} = \begin{cases} \frac{8e^A}{e^{2A}-2} & \text{for } W/d < 2 \\ \frac{2}{\pi}\left[B - 1 - \ln(2B-1) + \frac{\epsilon_r - 1}{2\epsilon_r}\left\{\ln(B-1) + 0.39 - \frac{0.61}{\epsilon_r}\right\}\right] & \text{for } W/d > 2 \end{cases}$$
- $$A = \frac{Z_o}{60}\sqrt{\frac{\epsilon_r+1}{2}} + \frac{\epsilon_r-1}{\epsilon_r+1}\left(0.23 + \frac{0.11}{\epsilon_r}\right)$$
- $$B = \frac{377\pi}{2Z_o\sqrt{\epsilon_r}}$$

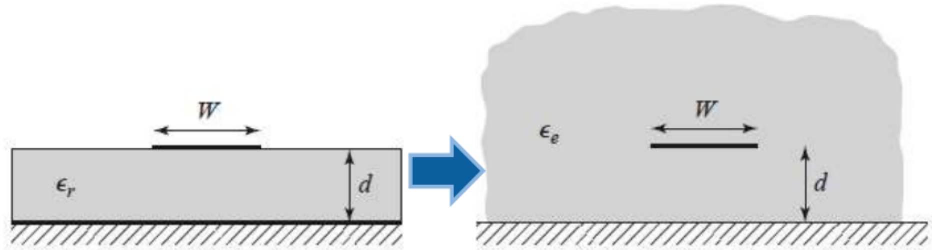


$$\cancel{A = 1.159 \rightarrow \frac{W}{d} = 3.126 \nless 2\ (rejected)}$$

$$B = 7.985 \rightarrow \frac{W}{d} = 3.081 > 2$$

Microstrip transmission lines: ϵ_e

The effective dielectric constant of a microstrip line is given approximately by: $\epsilon_e = \frac{\epsilon_r+1}{2} + \frac{\epsilon_r-1}{2}\frac{1}{\sqrt{1+12d/W}}$.

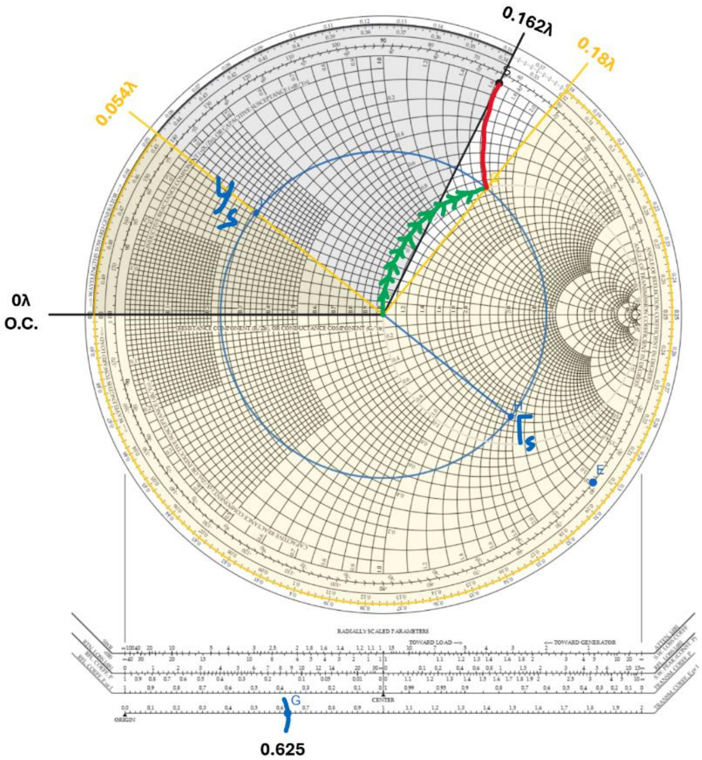


$$\varepsilon_{eff} = 1.871$$

$$\lambda = \lambda_g = \frac{\lambda_o}{\sqrt{\varepsilon_{eff}}} = \frac{c_o}{f\sqrt{\varepsilon_{eff}}} = 0.023069m$$

$$\lambda = 23.069mm \rightarrow \frac{\lambda}{4} = 5.767mm$$

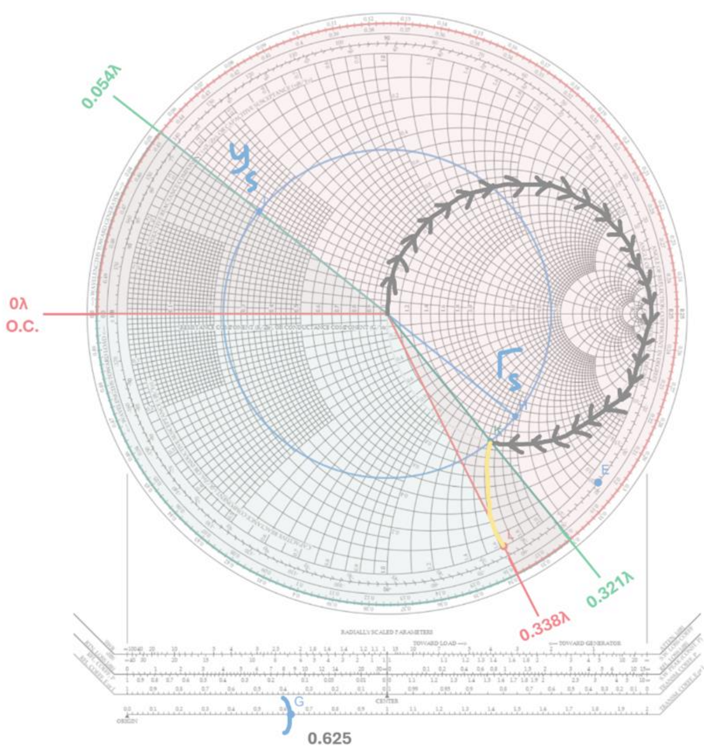
Input and Output Matching Network



Solution1

$$l = 0.162\lambda - 0\lambda$$
$$l = 0.162\lambda + \frac{m\lambda}{2}, m = 0, 1, 2, \ldots$$
$$l = 0.162\lambda = 58.32°$$
$$l = 3.737mm$$

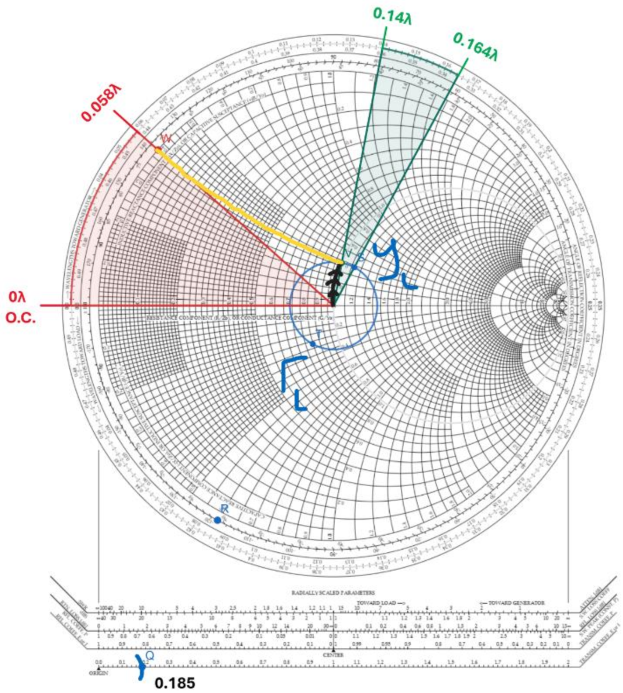
$$d = 0.054\lambda - 0.18\lambda$$
$$d = -0.126\lambda$$
$$d = -0.126\lambda + 0.5\lambda$$
$$d = 0.374\lambda + \frac{m\lambda}{2}, m = 0, 1, 2, \ldots$$
$$d = 0.374\lambda = 134.64°$$
$$d = 8.628mm$$



Solution2

$$l = 0.338\lambda - 0\lambda$$
$$l = 0.338\lambda + \frac{m\lambda}{2}, m = 0, 1, 2, \ldots$$
$$l = 0.338\lambda = 121.68°$$
$$l = 7.797mm$$

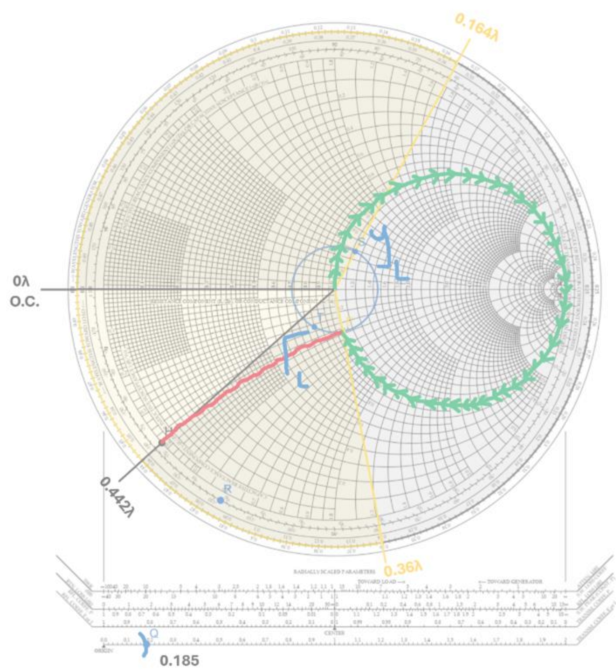
$$d = 0.054\lambda - 0.321\lambda$$
$$d = -0.267\lambda$$
$$d = -0.267\lambda + 0.5\lambda$$
$$d = 0.233\lambda$$
$$d = 0.233\lambda + \frac{m\lambda}{2}, m = 0, 1, 2, \ldots$$
$$d = 0.233\lambda = 83.88°$$
$$d = 5.375mm$$



Solution1

$$l = 0.058\lambda - 0\lambda$$
$$l = 0.058\lambda + \frac{m\lambda}{2}, m = 0, 1, 2, \ldots$$
$$l = 0.058\lambda = 20.88°$$
$$l = 1.338mm$$

$$d = 0.164\lambda - 0.14\lambda$$
$$d = 0.024\lambda + \frac{m\lambda}{2}, m = 0, 1, 2, \ldots$$
$$d = 0.024\lambda = 8.64°$$
$$d = 0.554mm$$

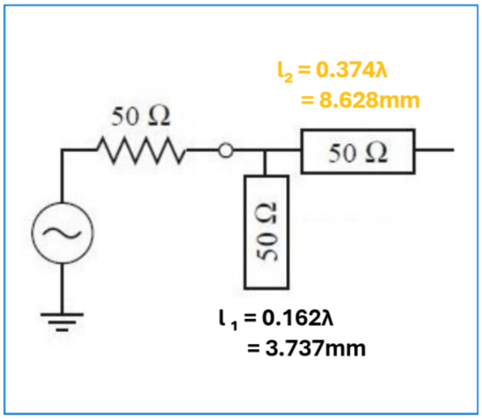


Solution2

$$l = 0.442\lambda - 0\lambda$$
$$l = 0.442\lambda + \frac{m\lambda}{2}, m = 0, 1, 2, \ldots$$
$$l = 0.442\lambda = 159.12°$$
$$l = 10.197mm$$

$$d = 0.164\lambda - 0.36\lambda$$
$$d = -0.196\lambda$$
$$d = -0.196\lambda + 0.5\lambda$$
$$d = 0.304\lambda + \frac{m\lambda}{2}, m = 0, 1, 2, \ldots$$
$$d = 0.304\lambda = 109.44°$$
$$d = 7.013mm$$

Input Matching Network



Output Matching Network